BMJ Open

BMJ Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjopen.bmj.com).

If you have any questions on BMJ Open's open peer review process please email info.bmjopen@bmj.com

BMJ Open

The impact of the COVID-19 pandemic on the mental health and learning of college and university students: a protocol of systematic review and meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-046428
Article Type:	Protocol
Date Submitted by the Author:	04-Nov-2020
Complete List of Authors:	Mao, Jing; Lanzhou University Second Hospital Gao, Xin; Lanzhou University Second Hospital Yan, Peireng; Geological Environmental Monitoring Institute of Gansu Province Ren, Xiaocao; Lanzhou University Second Hospital Guan, Yong; Lanzhou University, Yan, Yi; Lanzhou University Second Hospital,
Keywords:	COVID-19, MENTAL HEALTH, PSYCHIATRY, Child & adolescent psychiatry < PSYCHIATRY, Adult psychiatry < PSYCHIATRY, Public health < INFECTIOUS DISEASES

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

The impact of the COVID-19 pandemic on the mental health and learning of college and university students: a protocol of systematic review and meta-analysis

Jing Mao[1, §], Xin Gao[2, §], Peireng Yan[3, 4, 5, §], Xiaocao Ren[2], Yong Guan[6], Yi Yan[7, *]

1. Department of Mental Health, Lanzhou University Second Hospital, Lanzhou, 730000, P.R. China

2. The Second Clinical Medical College of Lanzhou University, Lanzhou, 730000, P.R. China

3. Geological Environmental Monitoring Institute of Gansu Province, Lanzhou, 730000, P.R. China

4. Enviromental Geology and Disaster Prevention and Control of Engineering Technology Research Center in Gansu Province, Lanzhou, 730000, P.R. China

5. Gansu Province Key Laboratory of Groundwater Engineering and Geothermal Resources, Lanzhou, 730000, P.R. China

6. The First Clinical Medical College of Lanzhou University, Lanzhou, 730000, P.R. China

7. Department of Project Management Office, Lanzhou University Second Hospital, Lanzhou, 730000, P.R. China

§ Jing Mao, Xin Gao and Peireng Yan are co-first author

*** Correspondence:**

Yi Yan

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Department of Project Management Office, Lanzhou University Second Hospital,

No. 82 Cuiyingmen, Lanzhou, Gansu, 730000, P.R. China

Tel: +86-15002665669

Fax: +86-0931- 8458109

Email: 15002665669@139.com

Running title: COVID-19 pandemic on mental health and learning of college and university students

Count:

Abstract: 259

Text: 1697

Number of figures: 0

Number of tables: 0

Conflicts of Interest

All authors have no conflicts of interest to declare**.**

Email from co-authors:

Jing Mao, 13919352350@139.com

Xin Gao, za123456zalh@163.com

Peireng Yan, yanpeireng@sina.com

Xiaocao Ren, renxc16@163.com

Yong Guan, yong__guan@163.com

For peer review only

Strengths and limitations of this study

- This study will be the first synthesis of research into the impact COVID-19 pandemic on the mental health and learning in college and university students.
- The findings from this study can be used to provide evidence for the field of pandemic and mental health and learning in college and university students.
- This study will explore the differences in the impact of the COVID-19 pandemic between various students.
- Heterogeneity in the health status and other characteristics of college and university students.
- Heterogeneity in the assessment of the mental health.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Abstract

Introduction: Coronavirus disease 2019 (COVID-19) has a serious impact on human physical health and mental health. The COVID-19 pandemic forced college and university students to take online classes, which may have bad impacts on students' learning. In addition, the student lost many job opportunities during the pandemic. Faced with employment and study pressure and worried about the epidemic, college and university students were prone to increase overall negative emotion, anxiety, and depression. Therefore, this systematic review and meta-analysis will be conducted to explore the impact of the COVID-19 pandemic on the mental health and learning of college and university students.

Methods and Analysis: We will conduct electronic literature search of the PubMed, Embase, Cochrane Library, Web of Science and Chinese National Knowledge Infrastructure (CNKI) databases. Two researchers will independently screen the studies, extract data, and assess the quality of the included study. Any disagreement will be resolved by the third investigator. The Newcastle-Ottawa Scale (NOS) and other tools will be used to assess the risk of bias, according to the study design of included studies. Odds ratio (OR), risk ratio (RR), mean difference, and 95% confidence interval will be considered as the effect size. Heterogeneity between studies will be assessed by subgroup and sensitivity analysis, and publication bias will be detected by funnel plots, Begg's test and Egger's test.

Ethics and Dissemination: This systematic review and meta-analysis involves no patient contact and no interaction with healthcare providers or systems. We will

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

disseminate the findings of this study through the presentation at scientific conferences and publication in a peer-reviewed journal.

Registration: The PROSPERO—International Prospective Register of Systematic Reviews—registration number is CRD (42020201132).

Keywords: COVID-19; Mental health; college and university students; learning; Systematic review; meta-analysis

Introduction

Coronavirus disease 2019 (COVID-19) is a newly discovered infectious disease caused by the SARS-CoV-2.[1] COVID-19 has been classified as a pandemic by the World Health Organization (WHO) on March 11, 2020.9.[2] WHO reported, as of 28 October 2020, there were almost 43.5 million confirmed COVID-19 cases, including more than 1 million deaths.[3] In the past two months, the number of new cases per week has remained more than 2 million.[4]

With various transmitted modes,[5] COVID-19 has a serious impact on human physical health. People infected this virus may be asymptomatic, or they may have six different "types" symptoms.[6] The mainly symptoms of COVID-19 were the respiratory and digestive system symptoms,[7] severe cases may lead to respiratory failure, cardiac arrest and death.[8]

Additionally, COVID-19 also has a serious impact on human mental health. During the COVID-19 pandemic, people suffer from the wider social impact of the pandemic, the governmental response, unemployment, financial difficulties, and other experiences such as concern about their illness, stigma, amnesia or traumatic

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

memories of severe illness.[9] Therefore, both the infected and non-infected population were under tremendous psychological pressure.[10]

College and university students should maintain a good attitude and work hard to learn knowledge, adapting to social life in advance, and striving for the future. However, the measures to prevent the further spread of COVID-19 made the lifestyle and learning of college and university students changed. And the changes made the student prone to anxiety and irritability.[11] [12] Moreover, some students were prone to have mental health problems because of the pressure of employment and entrance examinations.[13] A survey research into the mental health of college students during the COVID-19 pandemic found some students were depressed and those who suffered severely had a tendency to commit suicide.[14] Additionally, research found that those age 21-40 years old and with the higher education level were more likely to be anxious and depressed during the COVID-19 pandemic.[15]

On the other hand, the mental health of college and university students will affect the attention, energy, and motivation of students. And all these had vital impact on the college and university students' learning and ability to obtain good higher education.[16] [17]

Therefore, this systematic review and meta-analysis will be conducted to identify the impact on mental health and learning of college and university students during the COVID-19 pandemic.

Methods

The PROSPERO—International Prospective Register of Systematic

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Reviews—registration number is CRD (42020201132). The study will be conducted according to the Preferred Reporting Items for Systematic reviews and Meta-Analysis Protocols (PRISMA-P).[18]

Patient and Public Involvement

No patient involved. No requirement of ethical approval and informed consent is needed because it is a retrospective study.

2.1. Search strategy

We will conduct electronic literature search of the PubMed, Embase, Cochrane Library, Web of Science and Chinese National Knowledge Infrastructure (CNKI) databases. There are no time and languages restrictions. The search strategy will include key words relating to COVID-19 and mental health. The keywords will be used and a systematic literature search will be carried out based on the following keywords: ("Students"[Mesh]) AND (((((((((("severe acute respiratory syndrome coronavirus 2" [Supplementary Concept]) OR (2019-nCoV[Title/Abstract])) OR (SARS-CoV-2[Title/Abstract])) OR (2019 novel coronavirus[Title/Abstract])) OR (COVID-19 virus[Title/Abstract])) OR (coronavirus disease 2019 virus[Title/Abstract])) OR (COVID19 virus[Title/Abstract])) OR (("COVID-19" [Supplementary Concept]) OR ((((((((((2019 novel coronavirus disease[Title/Abstract]) OR (COVID19[Title/Abstract])) OR (COVID-19 pandemic[Title/Abstract])) OR (SARS-CoV-2 infection[Title/Abstract])) OR (COVID-19 virus disease[Title/Abstract])) OR (2019 novel coronavirus infection[Title/Abstract])) OR (2019-nCoV infection[Title/Abstract])) OR

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

(coronavirus disease 2019[Title/Abstract])) OR (coronavirus disease-19[Title/Abstract])) OR (2019-nCoV disease[Title/Abstract])) OR (COVID-19 virus infection[Title/Abstract]))). References of included study will also be traced back to find potential qualified studies.

2.2. Inclusion criteria

We will include all peer-reviewed articles exploring mental health and learning problems of college and university students during the COVID-19 pandemic.

Types of participants/populationCollege and university students exposed to the COVID-19 pandemic, with no age range, gender, ethnicity, or health status restrictions.

Types of study designs

We will include all peer-reviewed primary studies that use any study design (e.g. randomized trial, cohort, case-control, longitudinal, cross-sectional, etc.).

Types of outcomes

The mental health of college and university students during the COVID-19 pandemic (e.g. depression, anxiety, post-injury stress disorder, suicide or suicidal ideation, fatigue and laziness, and psychological interventions). We also choose the outcomes about the impact of the COVID-19 pandemic on the learning of college and university students.

2.3. Exclusion criteria

Studies will be excluded if one of the following conditions is met: 1) the research data are not complete; 2) qualitative studies, editors, letters, comments, notes, reviews, and

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

meta-analyses; 3) the study is not in Chinese or English.

2.4. Study selection

Two researchers will independently screen the studies retrieved during the searches to identify those that may meet the inclusion criteria, and to exclude irrelevant studies. Any conflicts arise during the screening of the articles will be resolved by the third researcher. Titles and abstracts of the studies retrieved by the literature search will be screened based on inclusion/exclusion criteria, and we will acquire the full text of potentially relevant studies for further assessment.

2.5. Data extraction

A standard form will be used to extract data from the included studies. Two reviewers will perform the data extraction independently. Discrepancies between the reviewers will be resolved either by discussion or by a third reviewer. To identify other relevant study data, we will contact the authors of published studies for incomplete data.

Information to be extracted include four domains: 1) study characteristics (authors, year of publication, country, date of data collection); 2) participant information (age, gender, ethnicity, health status, institutions of higher education, and other characteristics such as infected/not infected with COVID-19); (3) methods (study design, measures, validity, reliability); and 4) any other information including COVID-19 related information about the study that is regarded as necessary.

2.6. Risk of bias evaluation

Two researchers will independently assess the risk of bias of included studies. If there is any question, a third investigator will solve it through discussion. The

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Newcastle-Ottawa Scale (NOS)[19] and other tools will be used to assess the risk of bias, according to the study design of included studies.

2.7. Data synthesis and meta-analysis

The available evidence will be synthesized in a meta-analysis to derive a pool estimate of the problems of mental health and learning for college and university students during COVID-19. This is dependent on enough relevant studies for a meta-analysis.

In this meta-analysis, Odds ratio (OR), risk ratio (RR) and 95% confidence interval (CI) will be considered as the effect size for dichotomous outcomes, weighted mean difference (WMD) or standardized mean difference (SMD) with 95% CI will be calculated as the effect size for continuous outcomes. Forest plots will be produced to visually assess the effect size and corresponding 95% CI using random-effects models.

Heterogeneity between studies will be assessed via the forest plot, while I^2 values described the total variation between studies. I^2 values of <25%, 25%–50%, and >50% indicate low, moderate, and high heterogeneity, respectively. P values for heterogeneity were derived using the Cochran Q statistic.

Funnel plots, Begg's test and Egger's test[20] would be adopted to detect publication bias only when there are at least 10 studies reporting the primary outcomes. All statistical analyses were performed with STATA version 15.

Subgroup analysis and sensitivity analysis

We will perform subgroup analyses using the following group variables: 1) the health

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

of college and university students(college and university students infected by 2019-nCoV; identified as suspected patients or in close contact with 2019-nCoV patients; general college and university students, etc.); and 2) the type of college and university students: medical students, overseas students or exchange student, and other students.

Discussion

Facing the pressure of study, employment and worried about the epidemic of COVID-19, college and university students were prone to had negative emotions.[21] Moreover, college and university students were high-risk population in mental health for the age and higher education level.[22] The mental health problems and the COVID-19 pandemic had impact on the learning of college and university students. Therefore, we conducted this systematic review and meta-analysis to provide evidence on the impact of the COVID-19 pandemic on college and university students mental health and learning. We hope the results can provide evidence for current research and new research direction of the related fields.

However, there may be some limitations for this research. The mental health status of the students before the COVID-19 pandemic, economic status and other factors have influences on the mental health and learning during the pandemic, but we cannot do the subgroup analysis of these factors for the insufficient data.

Funding The authors have not declared any grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Competing interests None declared.

Patient consent for publication Not required.

Author contributions

Conceptualization: Yi Yan and Jing Mao

Data curation: Jing Mao, Xin Gao and Peireng Yan

Formal analysis: Peireng Yan and Xin Gao

Investigation: Jing Mao, Xin Gao, Peireng Yan and Yong Guan

Methodology: Yi Yan, Jing Mao, Xin Gao and Peireng Yan

Project administration: Yi Yan and Jing Mao

Software: Jing Mao, Xin Gao, Peireng Yan and Xiaocao Ren

Supervision: Yi Yan and Jing Mao

Writing – original draft: Jing Mao, Xin Gao and Peireng Yan

Writing – review and editing: Yi Yan, Jing Mao, Xin Gao, Peireng Yan, Xiaocao Ren and Yong Guan

References

1. Dömling A, Gao L. Chemistry and Biology of SARS-CoV-2. *Chem* 2020;6:1283-95.
2. Espinosa OA, Zanetti ADS, Antunes EF, et al. Prevalence of comorbidities in patients and mortality cases affected by SARS-CoV2: a systematic review and meta-analysis. *Rev Inst Med Trop Sao Paulo* 2020;62:e43.
3. Organization WH. Coronavirus disease (COVID-19) pandemic: World Health Organization; 2020 [Available from: https://www.who.int/emergencies/diseases/novel-coronavirus-2019 accessed 28, Oct 2020.
4. Organization WH. WHO Coronavirus Disease (COVID-19) Dashboard: World Health Organization; 2020 [Available from: https://covid19.who.int/ accessed 28, Oct 2020.
5. Yu IT, Li Y, Wong TW, et al. Evidence of airborne transmission of the severe acute respiratory syndrome virus. *N Engl J Med* 2004;350:1731-9.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

6. Wise J. Covid-19: Study reveals six clusters of symptoms that could be used as a clinical prediction tool. *Bmj* 2020;370:m2911.
7. Chen N, Zhou M, Dong X, et al. Epidemiological and clinical characteristics of 99 cases of 2019 novel coronavirus pneumonia in Wuhan, China: a descriptive study. *Lancet* 2020;395:507-13.
8. Denault AY, Delisle S, Canty D, et al. A proposed lung ultrasound and phenotypic algorithm for the care of COVID-19 patients with acute respiratory failure. *Can J Anaesth* 2020;67:1393-404.
9. Rogers JP, Chesney E, Oliver D, et al. Psychiatric and neuropsychiatric presentations associated with severe coronavirus infections: a systematic review and meta-analysis with comparison to the COVID-19 pandemic. *Lancet Psychiatry* 2020;7:611-27.
10. Rehman S, Lela U. Psychological Aid to COVID-19 Pandemic: A Mental Health Response to Crises Management. *Psychiatr Danub* 2020;32:262-65.
11. Walke HT, Honein MA, Redfield RR. Preventing and Responding to COVID-19 on College Campuses. *Jama* 2020
12. Wilson E, Donovan CV, Campbell M, et al. Multiple COVID-19 Clusters on a University Campus - North Carolina, August 2020. *MMWR Morb Mortal Wkly Rep* 2020;69:1416-18.
13. Zhang Y, Zhang H, Ma X, et al. Mental Health Problems during the COVID-19 Pandemics and the Mitigation Effects of Exercise: A Longitudinal Study of College Students in China. *Int J Environ Res Public Health* 2020;17
14. Wang X, Hegde S, Son C, et al. Investigating Mental Health of US College Students During the COVID-19 Pandemic: Cross-Sectional Survey Study. *J Med Internet Res* 2020;22:e22817.
15. Zheng Z, Peng F, Xu B, et al. Risk factors of critical & mortal COVID-19 cases: A systematic literature review and meta-analysis. *J Infect* 2020;81:e16-e25.
16. Arenas DL, Viduani AC, Bassols AMS, et al. Peer support intervention as a tool to address college students' mental health amidst the COVID-19 pandemic. *Int J Soc Psychiatry* 2020:20764020954468.
17. Andrews JL, Foulkes L, Blakemore SJ. Peer Influence in Adolescence: Public-Health Implications for COVID-19. *Trends Cogn Sci* 2020;24:585-87.
18. Moher D, Shamseer L, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst Rev* 2015;4:1.
19. GA Wells BS, D O'Connell, J Peterson, V Welch, M Losos, P Tugwell. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses 2019 [Available from: http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp accessed 23 Oct 2020.
20. Irwig L, Macaskill P, Berry G, et al. Bias in meta-analysis detected by a simple, graphical test. Graphical test is itself biased. *Bmj* 1998;316:470; author reply 70-1.
21. Liang L, Ren H, Cao R, et al. The Effect of COVID-19 on Youth Mental Health. *Psychiatr Q* 2020;91:841-52.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

22. Son C, Hegde S, Smith A, et al. Effects of COVID-19 on College Students' Mental Health in the United States: Interview Survey Study. *J Med Internet Res* 2020;22:e21279.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol

Section and topic	Item No	Checklist item	No. Page
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	NA
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	4
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1-2
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	12
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	NA
Support:			
Sources	5a	Indicate sources of financial or other support for the review	11
Sponsor	5b	Provide name for the review funder and/or sponsor	NA
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	NA
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	5-6
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7-9
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	7-8
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	9

For peer review only

Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	9
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	9-10
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	9
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	8
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9-10
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	10
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	10
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	10-11
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	NA
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	10
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	6-7

NA, not applicable

For peer review only

BMJ Open

The impact of the COVID-19 pandemic on the mental health and learning of college and university students: a protocol of systematic review and meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-046428.R1
Article Type:	Protocol
Date Submitted by the Author:	10-Mar-2021
Complete List of Authors:	Mao, Jing; Lanzhou University Second Hospital Gao, Xin; Lanzhou University Second Hospital Yan, Peireng; Geological Environmental Monitoring Institute of Gansu Province; Enviromental Geology and Disaster Prevention and Control of Engineering Technology Research Center in Gansu Province Ren, Xiaocao; Lanzhou University Second Hospital Guan, Yong; Lanzhou University, Yan, Yi; Lanzhou University Second Hospital,
Primary Subject Heading:	Mental health
Secondary Subject Heading:	Public health
Keywords:	COVID-19, MENTAL HEALTH, PSYCHIATRY, Child & adolescent psychiatry < PSYCHIATRY, Adult psychiatry < PSYCHIATRY, Public health < INFECTIOUS DISEASES

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

The impact of the COVID-19 pandemic on the mental health and learning of college and university students: a protocol of systematic review and meta-analysis

Jing Mao[1, §], Xin Gao[2, §], Peireng Yan[3, 4, 5, §], Xiaocao Ren[2], Yong Guan[6], Yi Yan[7, *]

1. Department of Mental Health, Lanzhou University Second Hospital, Lanzhou, 730000, P.R. China

2. The Second Clinical Medical College of Lanzhou University, Lanzhou, 730000, P.R. China

3. Geological Environmental Monitoring Institute of Gansu Province, Lanzhou, 730000, P.R. China

4. Enviromental Geology and Disaster Prevention and Control of Engineering Technology Research Center in Gansu Province, Lanzhou, 730000, P.R. China

5. Gansu Province Key Laboratory of Groundwater Engineering and Geothermal Resources, Lanzhou, 730000, P.R. China

6. The First Clinical Medical College of Lanzhou University, Lanzhou, 730000, P.R. China

7. Department of Project Management Office, Lanzhou University Second Hospital, Lanzhou, 730000, P.R. China

§ Jing Mao, Xin Gao and Peireng Yan are co-first author

*** Correspondence:**

Yi Yan

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Department of Project Management Office, Lanzhou University Second Hospital,

No. 82 Cuiyingmen, Lanzhou, Gansu, 730000, P.R. China

Tel: +86-15002665669

Fax: +86-0931- 8458109

Email: 15002665669@139.com

Running title: COVID-19 pandemic on mental health and learning of college and university students

Count:

Abstract: 259

Text: 2027

Number of figures: 0

Number of tables: 0

Conflicts of Interest

All authors have no conflicts of interest to declare**.**

Email from co-authors:

Jing Mao, 13919352350@139.com

Xin Gao, za123456zalh@163.com

Peireng Yan, yanpeireng@sina.com

Xiaocao Ren, renxc16@163.com

Yong Guan, yong__guan@163.com

For peer review only

Abstract

Introduction: Coronavirus disease 2019 (COVID-19) has a serious impact on human physical health and mental health. The COVID-19 pandemic forced college and university students to take online classes, which may have bad impacts on students' learning. In addition, the student lost many job opportunities during the pandemic. Faced with employment and study pressure and worried about the epidemic, college and university students were prone to increase overall negative emotion, anxiety, and depression. Therefore, this systematic review and meta-analysis will be conducted to explore the impact of the COVID-19 pandemic on the mental health and learning of college and university students.

Methods and Analysis: We will conduct electronic literature search of the PubMed, Embase, Cochrane Library, Web of Science and Chinese National Knowledge Infrastructure (CNKI) databases. Two researchers will independently screen the studies, extract data, and assess the quality of the included study. Any disagreement will be resolved by the third investigator. The Newcastle-Ottawa Scale (NOS) and other tools will be used to assess the risk of bias, according to the study design of included studies. Odds ratio (OR), risk ratio (RR), mean difference, and 95% confidence interval will be considered as the effect size. Heterogeneity between studies will be assessed by subgroup and sensitivity analysis, and publication bias will be detected by funnel plots, Begg's test and Egger's test.

Ethics and Dissemination: This systematic review and meta-analysis involves no patient contact and no interaction with healthcare providers or systems. We will

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

disseminate the findings of this study through the presentation at scientific conferences and publication in a peer-reviewed journal.

Registration: The PROSPERO—International Prospective Register of Systematic Reviews—registration number is CRD (42020201132).

Keywords: COVID-19; Mental health; college and university students; learning; Systematic review; meta-analysis

For peer review only

Strengths and limitations of this study

- This study will be the first synthesis of research into the impact COVID-19 pandemic on the mental health and learning in college and university students.
- The study selection, data extraction and risk of bias evaluation will be performed by two independent reviewers to minimize the potential bias.
- This study will explore the differences in the impact of the COVID-19 pandemic between various students.
- Only the studies published in English or Chinese will be included because of the limited resource and the authors' linguistic competence.
- The different assessment scales of the mental health will be a major source of heterogeneity for this study.

Introduction

Coronavirus disease 2019 (COVID-19) is a newly discovered infectious disease caused by the SARS-CoV-2.[1] COVID-19 has been classified as a pandemic by the

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

World Health Organization (WHO) on March 11, 2020.9.[2] WHO reported, as of 28 October 2020, there were almost 43.5 million confirmed COVID-19 cases, including more than 1 million deaths.[3] In the past two months, the number of new cases per week has remained more than 2 million.[4]

With various transmitted modes,[5] COVID-19 has a serious impact on human physical health. People infected this virus may be asymptomatic, or they may have six different "types" symptoms.[6] The mainly symptoms of COVID-19 were the respiratory and digestive system symptoms,[7] severe cases may lead to respiratory failure, cardiac arrest and death.[8]

Additionally, COVID-19 also has a serious impact on human mental health. During the COVID-19 pandemic, people suffer from the wider social impact of the pandemic, the governmental response, unemployment, financial difficulties, and other experiences such as concern about their illness, stigma, amnesia or traumatic memories of severe illness.[9] Several systematic review and meta-analyses suggesting an important impact of the COVID-19 outbreak on general population's mental health [9-12] and in specific population,[13-15] Therefore, both the infected and non-infected population were under tremendous psychological pressure.[16]

College and university students should maintain a good attitude and work hard to learn knowledge, adapting to social life in advance, and striving for the future. However, the measures to prevent the further spread of COVID-19 made the lifestyle and learning of college and university students changed. And the changes made the student prone to anxiety and irritability.[17 18] Moreover, some students were prone to

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

have mental health problems because of the pressure of employment and entrance examinations.[19] Additionally, research found that those age 21-40 years old and with the higher education level were more likely to be anxious and depressed during the COVID-19 pandemic,[20 21] Recently, several surveies research into the mental health of college students during the COVID-19 pandemic found there is a high prevalence of anxiety, depression and other mental health among university and college students,[22-25] they even at higher risk of experiencing mental health problems than other populations.[26 27]

On the other hand, the mental health of college and university students will affect the attention, energy, and motivation of students. And all these had vital impact on the college and university students' learning and ability to obtain good higher education.[28 29]

High quality Meta-anlysis has been increasingly regarded as one of the key tools for achieving evidence. But there was no systematic review and meta-analysis on the impact of the COVID-19 pandemic on the mental health and learning of college and university students. Therefore, this systematic review and meta-analysis will be conducted to identify the impact on mental health and learning of college and university students during the COVID-19 pandemic.

Review question

What is the impact of the COVID-19 pandemic on the mental health and learning of college and university students?

Methods

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

The PROSPERO—International Prospective Register of Systematic Reviews—registration number is CRD (42020201132). The Preferred Reporting Items for Systematic reviews and Meta-Analysis Protocols (PRISMA-P) guideline were utilized to format this protocol.[30]

Patient and Public Involvement

No patient involved. No requirement of ethical approval and informed consent is needed because it is a retrospective study.

Search strategy

We will conduct electronic literature search of the PubMed, Embase, Cochrane Library, Web of Science and Chinese National Knowledge Infrastructure (CNKI) databases. There are no time and languages restrictions. The search strategy will include key words relating to COVID-19 and mental health. The keywords will be used and a systematic literature search will be carried out based on the following keywords: ("Students"[Mesh]) AND ((((((((("severe acute respiratory syndrome coronavirus 2" [Supplementary Concept]) OR (2019-nCoV[Title/Abstract])) OR (SARS-CoV-2[Title/Abstract])) OR (2019 novel coronavirus[Title/Abstract])) OR (COVID-19 virus[Title/Abstract])) OR (coronavirus disease 2019 virus[Title/Abstract])) OR (COVID19 virus[Title/Abstract])) OR (("COVID-19" [Supplementary Concept]) OR ((((((((((2019 novel coronavirus disease[Title/Abstract]) OR (COVID19[Title/Abstract])) OR (COVID-19 pandemic[Title/Abstract])) OR (SARS-CoV-2 infection[Title/Abstract])) OR (COVID-19 virus disease[Title/Abstract])) OR (2019 novel coronavirus

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

infection[Title/Abstract])) OR (2019-nCoV infection[Title/Abstract])) OR (coronavirus disease 2019[Title/Abstract])) OR (coronavirus disease-19[Title/Abstract])) OR (2019-nCoV disease[Title/Abstract])) OR (COVID-19 virus infection[Title/Abstract]))). References of included study will also be traced back to find potential qualified studies.

Inclusion criteria

We will include all peer-reviewed articles exploring mental health and learning problems of college and university students during the COVID-19 pandemic.

Types of participants/population

College and university students exposed to the COVID-19 pandemic, with no age range, gender, ethnicity, or health status restrictions.

Types of study designs

We will include all peer-reviewed primary studies that use any study design (e.g. randomized trial, cohort, case-control, longitudinal, cross-sectional, etc.).

Types of outcomes

The mental health of college and university students during the COVID-19 pandemic (e.g. depression, anxiety, post-injury stress disorder, suicide or suicidal ideation, fatigue and laziness, and psychological interventions). We also choose the outcomes about the impact of the COVID-19 pandemic on the learning of college and university students.

Exclusion criteria

Studies will be excluded if one of the following conditions is met: 1) the research data

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

are not complete; 2) qualitative studies, editors, letters, comments, notes, reviews, and meta-analyses; 3) the study is not in Chinese or English.

Study selection

Two researchers will independently screen the studies retrieved during the searches to identify those that may meet the inclusion criteria, and to exclude irrelevant studies. Any conflicts arise during the screening of the articles will be resolved by the third researcher. Titles and abstracts of the studies retrieved by the literature search will be screened based on inclusion/exclusion criteria, and we will acquire the full text of potentially relevant studies for further assessment.

Data extraction

A standard form will be used to extract data from the included studies. Two reviewers will perform the data extraction independently. Discrepancies between the reviewers will be resolved either by discussion or by a third reviewer. To identify other relevant study data, we will contact the authors of published studies for incomplete data.

Information to be extracted include four domains: 1) study characteristics (authors, year of publication, country, date of data collection); 2) participant information (age, gender, ethnicity, health status, institutions of higher education, and other characteristics such as infected/not infected with COVID-19); (3) methods (study design, measures, validity, reliability); and 4) any other information including COVID-19 related information about the study that is regarded as necessary.

Risk of bias evaluation

Two researchers will independently assess the risk of bias of included studies. If there

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

is any question, a third investigator will solve it through discussion. The Newcastle-Ottawa Scale (NOS)[31] and other tools will be used to assess the risk of bias, according to the study design of included studies.

Data synthesis and meta-analysis

The available evidence will be synthesized in a meta-analysis to derive a pool estimate of the problems of mental health and learning for college and university students during COVID-19. This is dependent on enough relevant studies for a meta-analysis.

In this meta-analysis, Odds ratio (OR), risk ratio (RR) and 95% confidence interval (CI) will be considered as the effect size for dichotomous outcomes, weighted mean difference (WMD) or standardized mean difference (SMD) with 95% CI will be calculated as the effect size for continuous outcomes. Forest plots will be produced to visually assess the effect size and corresponding 95% CI using random-effects models.

Heterogeneity between studies will be assessed via the forest plot, while I^2 values described the total variation between studies. I^2 values of <25%, 25%–50%, and >50% indicate low, moderate, and high heterogeneity, respectively. P values for heterogeneity were derived using the Cochran Q statistic.

Funnel plots, Begg's test and Egger's test[32] would be adopted to detect publication bias only when there are at least 10 studies reporting the primary outcomes. All statistical analyses were performed with STATA version 15.

We will perform subgroup analyses using the following group variables: 1) the health

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

of college and university students (college and university students infected by 2019-nCoV; identified as suspected patients or in close contact with 2019-nCoV patients; general college and university students, etc.); and 2) the type of college and university students: medical students, overseas students or exchange student, and other students. We will carry out a sensitivity analysis by excluding one study each time one to explore whether the results were driven by a study with an extreme result. We will also perform meta-regression to explore the sources of heterogeneity when there are not fewer than 10 studies in a meta-analysis. [33]

Current status and plan

The review is undaring preliminary searches, and is expected to be completed on January 2022.

Ethics and Dissemination

This systematic review and meta-analysis involves no patient contact and no interaction with healthcare providers or systems. We will disseminate the findings of this study through the presentation at scientific conferences and publication in a peer-reviewed journal.

Discussion

COVID-19 also has a serious impact on human physical health. Several systematic review and meta-analyses suggesting the COVID-19 outbreak had an important impact on human mental health.[9-15] Facing the pressure of study, employment, great changes of life and worried about the epidemic of COVID-19, college and university

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

students were prone to had negative emotions.[34] Moreover, college and university students were high-risk population in mental health for the age and higher education level.[35] The mental health problems and the COVID-19 pandemic had impact on the learning of college and university students. We conducted this systematic review and meta-analysis to provide evidence on the impact of the COVID-19 pandemic on college and university students mental health and learning.

However, there may be some limitations for this research. Firstly, the mental health status of the students before the COVID-19 pandemic, economic status and other factors have influences on the mental health and learning during the pandemic, but we cannot do the subgroup analysis of these factors for the insufficient data. Secondly, Only the studies published in English or Chinese will be included because of the limited resource and the authors' linguistic competence. Additionally, the use of different scales to assess mental health maybe a major source of the heterogeneity in this study.

In summary, this systematic review and meta-analysis will provide evidence on the impact of the COVID-19 pandemic on the mental health and learning of college and university students. The final report of this study in the form of scientific paper will be published in peer-reviewed journal. It is expected that the findings of this review provide evidence for psychiatrists/psychologist, public health and education specialists, policy makers and others working with public health emergencies.

Funding This study received funding from the science and technology department of Gansu Province, Project Grant Number: 20CX9ZA113.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Competing interests None declared.

Patient consent for publication Not required.

Author contributions

Conceptualization: Yi Yan and Jing Mao

Data curation: Jing Mao, Xin Gao and Peireng Yan

Formal analysis: Peireng Yan and Xin Gao

Investigation: Jing Mao, Xin Gao, Peireng Yan and Yong Guan

Methodology: Yi Yan, Jing Mao, Xin Gao and Peireng Yan

Project administration: Yi Yan and Jing Mao

Software: Jing Mao, Xin Gao, Peireng Yan and Xiaocao Ren

Supervision: Yi Yan and Jing Mao

Writing – original draft: Jing Mao, Xin Gao and Peireng Yan

Writing – review and editing: Yi Yan, Jing Mao, Xin Gao, Peireng Yan, Xiaocao Ren and Yong Guan

Ethical Approval Statement Not applicable.

References

1 Dömling A, Gao L. Chemistry and Biology of SARS-CoV-2. *Chem* 2020;6:1283-95.

2 Espinosa OA, Zanetti ADS, Antunes EF, et al. Prevalence of comorbidities in patients and mortality cases affected by SARS-CoV2: a systematic review and meta-analysis. *Rev Inst Med Trop Sao Paulo* 2020;62:e43.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

3 Organization WH. Coronavirus disease (COVID-19) pandemic: World Health Organization; 2020 [Available from: https://www.who.int/emergencies/diseases/novel-coronavirus-2019 accessed 27, Oct 2020.

4 Organization WH. WHO Coronavirus Disease (COVID-19) Dashboard: World Health Organization; 2020 [Available from: https://covid19.who.int/ accessed 27, Oct 2020.

5 Yu IT, Li Y, Wong TW, et al. Evidence of airborne transmission of the severe acute respiratory syndrome virus. *N Engl J Med* 2004;350:1731-9.

6 Wise J. Covid-19: Study reveals six clusters of symptoms that could be used as a clinical prediction tool. *Bmj* 2020;370:m2911.

7 Chen N, Zhou M, Dong X, et al. Epidemiological and clinical characteristics of 99 cases of 2019 novel coronavirus pneumonia in Wuhan, China: a descriptive study. *Lancet* 2020;395:507-13.

8 Denault AY, Delisle S, Canty D, et al. A proposed lung ultrasound and phenotypic algorithm for the care of COVID-19 patients with acute respiratory failure. *Can J Anaesth* 2020;67:1393-404.

9 Rogers JP, Chesney E, Oliver D, et al. Psychiatric and neuropsychiatric presentations associated with severe coronavirus infections: a systematic review and meta-analysis with comparison to the COVID-19 pandemic. *Lancet Psychiatry* 2020;7:611-27.

10 Bueno-Notivol J, Gracia-García P, Olaya B, et al. Prevalence of depression during

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

the COVID-19 outbreak: A meta-analysis of community-based studies. *International journal of clinical and health psychology : IJCHP* 2021;21:100196.

11 Santabárbara J, Lasheras I, Lipnicki DM, et al. Prevalence of anxiety in the COVID-19 pandemic: An updated meta-analysis of community-based studies. *Progress in neuro-psychopharmacology & biological psychiatry* 2020;109:110207.

12 Hessami K, Romanelli C, Chiurazzi M, et al. COVID-19 pandemic and maternal mental health: a systematic review and meta-analysis. *The journal of maternal-fetal & neonatal medicine : the official journal of the European Association of Perinatal Medicine, the Federation of Asia and Oceania Perinatal Societies, the International Society of Perinatal Obstet* 2020:1-8.

13 Yan H, Ding Y, Guo W. Mental Health of Pregnant and Postpartum Women During the Coronavirus Disease 2019 Pandemic: A Systematic Review and Meta-Analysis. *Frontiers in psychology* 2020;11:617001.

14 Salazar de Pablo G, Vaquerizo-Serrano J, Catalan A, et al. Impact of coronavirus syndromes on physical and mental health of health care workers: Systematic review and meta-analysis. *Journal of affective disorders* 2020;275:48-57.

15 Lasheras I, Gracia-García P, Lipnicki DM, et al. Prevalence of Anxiety in Medical Students during the COVID-19 Pandemic: A Rapid Systematic Review with Meta-Analysis. *International journal of environmental research and public health* 2020;17

16 Rehman S, Lela U. Psychological Aid to COVID-19 Pandemic: A Mental Health Response to Crises Management. *Psychiatr Danub* 2020;32:262-65.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

17 Walke HT, Honein MA, Redfield RR. Preventing and Responding to COVID-19 on College Campuses. *Jama* 2020

18 Wilson E, Donovan CV, Campbell M, et al. Multiple COVID-19 Clusters on a University Campus - North Carolina, August 2020. *MMWR Morb Mortal Wkly Rep* 2020;69:1416-18.

19 Zhang Y, Zhang H, Ma X, et al. Mental Health Problems during the COVID-19 Pandemics and the Mitigation Effects of Exercise: A Longitudinal Study of College Students in China. *International journal of environmental research and public health* 2020;17

20 Zheng Z, Peng F, Xu B, et al. Risk factors of critical & mortal COVID-19 cases: A systematic literature review and meta-analysis. *J Infect* 2020;81:e16-e25.

21 Ahmed MZ, Ahmed O, Aibao Z, et al. Epidemic of COVID-19 in China and associated Psychological Problems. *Asian journal of psychiatry* 2020;51:102092.

22 Wang X, Hegde S, Son C, et al. Investigating Mental Health of US College Students During the COVID-19 Pandemic: Cross-Sectional Survey Study. *J Med Internet Res* 2020;22:e22817.

23 Dhar BK, Ayittey FK, Sarkar SM. Impact of COVID-19 on Psychology among the University Students. *Global challenges (Hoboken, NJ)* 2020;4:2000038.

24 Debowska A, Horeczy B, Boduszek D, et al. A repeated cross-sectional survey assessing university students' stress, depression, anxiety, and suicidality in the early stages of the COVID-19 pandemic in Poland. *Psychological medicine* 2020:1-4.

25 Ghazawy ER, Ewis AA, Mahfouz EM, et al. Psychological impacts of COVID-19

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

pandemic on the university students in Egypt. *Health promotion international* 2020

26 Rogowska AM, Pavlova I, Kuśnierz C, et al. Does Physical Activity Matter for the Mental Health of University Students during the COVID-19 Pandemic? *Journal of clinical medicine* 2020;9

27 Solomou I, Constantinidou F. Prevalence and Predictors of Anxiety and Depression Symptoms during the COVID-19 Pandemic and Compliance with Precautionary Measures: Age and Sex Matter. *International journal of environmental research and public health* 2020;17

28 Arenas DL, Viduani AC, Bassols AMS, et al. Peer support intervention as a tool to address college students' mental health amidst the COVID-19 pandemic. *Int J Soc Psychiatry* 2020:20764020954468.

29 Andrews JL, Foulkes L, Blakemore SJ. Peer Influence in Adolescence: Public-Health Implications for COVID-19. *Trends Cogn Sci* 2020;24:585-87.

30 Moher D, Shamseer L, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst Rev* 2015;4:1.

31 GA Wells BS, D O'Connell, J Peterson, V Welch, M Losos, P Tugwell. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses 2019 [Available from: http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp accessed 23 Oct 2020.

32 Irwig L, Macaskill P, Berry G, et al. Bias in meta-analysis detected by a simple,

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

graphical test. Graphical test is itself biased. *Bmj* 1998;316:470; author reply 70-1.

33 Deeks JJ, Higgins JP, Altman DG. Cochrane Handbook for Systematic Reviews of Interventions. Version 6.2 [updated 2021].2021.

34 Liang L, Ren H, Cao R, et al. The Effect of COVID-19 on Youth Mental Health. *Psychiatr Q* 2020;91:841-52.

35 Son C, Hegde S, Smith A, et al. Effects of COVID-19 on College Students' Mental Health in the United States: Interview Survey Study. *J Med Internet Res* 2020;22:e21279.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol

Section and topic	Item No	Checklist item	No. Page
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	NA
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	4
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1-2
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	13
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	NA
Support:			
Sources	5a	Indicate sources of financial or other support for the review	13
Sponsor	5b	Provide name for the review funder and/or sponsor	13
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	NA
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	5-7
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	7
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7-9
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	7-8
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	9

For peer review only

Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	9
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	9-10
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	9
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	9
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	10
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	10
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	10-11
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	10-11
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	NA
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	11
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	7

NA, not applicable

For peer review only